Operating Systems
Week 6
Lecture/Discussion 6

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Previous

- Scheduling Algorithms (SJF Preemptive- Round Robin)

Today Discussion

- Continue Scheduling Algorithms
- Priority Algorithms

Priority Scheduling

A priority number (integer) is associated with each process

The CPU is allocated to the process with the highest priority (smallest integer ≡ highest priority)

- Preemptive
- Nonpreemptive

SJF is priority scheduling where priority is the inverse of predicted next CPU burst time

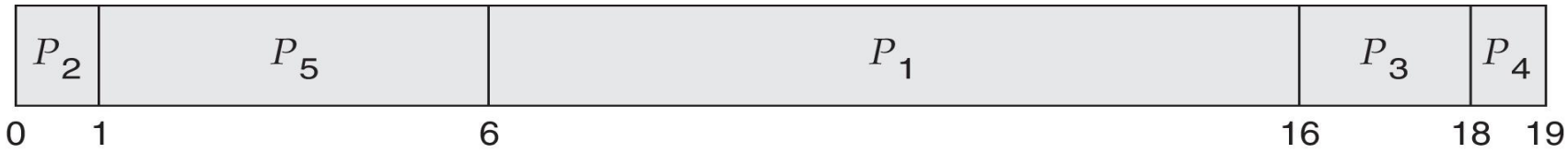
Problem ≡ **Starvation** – low priority processes may never execute

Solution ≡ **Aging** – as time progresses increase the priority of the process

Example of Priority Scheduling

Process	Burst Time	Priority
P_1	10	3
P_2	1	1
P_3	2	4
P_4	1	5
P_5	5	2

Priority scheduling Gantt Chart



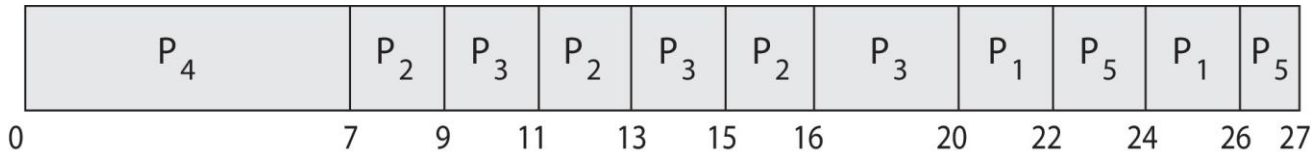
Average waiting time = 8.2

Priority Scheduling w/ Round-Robin

Run the process with the highest priority. Processes with the same priority run round-robin
Example:

Process	Burst Time	Priority
P_1	4	3
P_2	5	2
P_3	8	2
P_4	7	1
P_5	3	3

Gantt Chart with time quantum = 2



Multilevel Queue

The ready queue consists of multiple queues
Multilevel queue scheduler defined by the following parameters:

- Number of queues
- Scheduling algorithms for each queue
- Method used to determine which queue a process will enter when that process needs service
- Scheduling among the queues

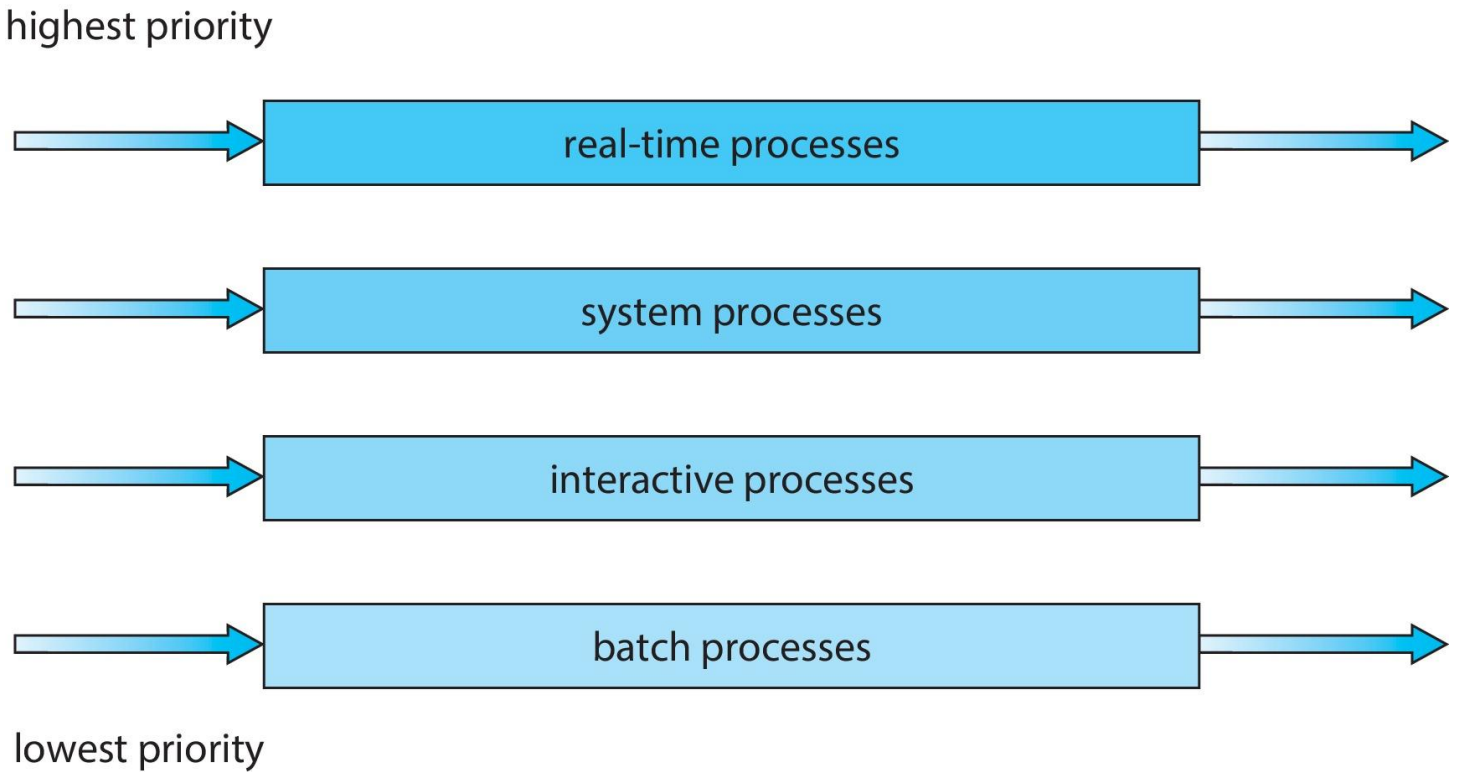
Multilevel Queue

With priority scheduling, have separate queues for each priority.
Schedule the process in the highest-priority queue!



Multilevel Queue

Prioritization based upon process type



Multilevel Feedback Queue

A process can move between the various queues.

Multilevel-feedback-queue scheduler defined by the following parameters:

- Number of queues
- Scheduling algorithms for each queue
- Method used to determine when to upgrade a process
- Method used to determine when to demote a process
- Method used to determine which queue a process will enter when that process needs service

Aging can be implemented using multilevel feedback queue

Example of Multilevel Feedback Queue

Three queues:

- Q_0 – RR with time quantum 8 milliseconds
- Q_1 – RR time quantum 16 milliseconds
- Q_2 – FCFS

Scheduling

- A new process enters queue Q_0 which is served in RR
  - When it gains CPU, the process receives 8 milliseconds
  - If it does not finish in 8 milliseconds, the process is moved to queue Q_1
- At Q_1 job is again served in RR and receives 16 additional milliseconds
  - If it still does not complete, it is preempted and moved to queue Q_2

